Example 9.6

Test the existence of the limit, $\lim_{x\to 1}\frac{4|x-1|+x-1}{|x-1|}$, $x\neq 1$.

Solution

For $x > 1$, $|x-1| = x-1$ and $f(1^+) = \lim_{x\to 1^+}\frac{4(x-1)+x-1}{x-1} = \lim_{x\to 1^+}\frac{5(x-1)}{(x-1)} = 5$.

For $x < 1$, $|x-1| = -(x-1)$, and $f(1^-) = \lim_{x\to 1^-}\frac{-4(x-1)+(x-1)}{-(x-1)} = \lim_{x\to 1^-}\frac{3(x-1)}{(x-1)} = 3$.

Thus $f(1^-) \neq f(1^+)$ and hence the limit does not exist.

EXERCISE 9.1

In problems 1-6, using the table estimate the value of the limit.

(1) $\lim_{x\to 2}\frac{x-2}{x^2-x-2}$

x	1.9	1.99	1.999	2.001	2.01	2.1
$f(x)$	0.344820	0.33444	0.33344	0.333222	0.33222	0.332258

(2) $\lim_{x\to 2}\frac{x-2}{x^2-4}$

x	1.9	1.99	1.999	2.001	2.01	2.1
$f(x)$	0.25641	0.25062	0.250062	0.24993	0.24937	0.24390

(3) $\lim_{x\to 0}\frac{\sqrt{x+3}-\sqrt{3}}{x}$

x	– 0.1	– 0.01	– 0.001	0.001	0.01	0.1
f(x)	0.2911	0.2891	0.2886	0.2886	0.2885	0.28631

(4) $\lim_{x\to -3}\frac{\sqrt{1-x}-2}{x+3}$

x	– 3.1	– 3.01	– 3.00	– 2.999	– 2.99	– 2.9
$f(x)$	– 0.24845	– 0.24984	– 0.24998	– 0.25001	– 0.25015	– 0.25158

(5) $\lim_{x\to 0}\frac{\sin x}{x}$

x	– 0.1	– 0.01	– 0.001	0.001	0.01	0.1
$f(x)$	0.99833	0.99998	0.99999	0.99999	0.99998	0.99833

(6) $\lim_{x\to 0}\frac{\cos x-1}{x}$

x	– 0.1	– 0.01	– 0.001	0.0001	0.01	0.1
$f(x)$	0.04995	0.0049999	0.0004999	–0.0004999	–0.004999	–0.04995

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